

Plan for NFL Distance Analysis

Goal

The goal of this assignment is to create a reproducible data analysis document using Quarto. This project investigates whether travel distance affects away team performance in the NFL by combining game-level performance metrics with calculated travel distances and presenting the results using tables and visualizations.

Needs

To complete this assignment, I will rely on Quarto and RStudio Desktop to build a reproducible PDF analysis document. I will use NFL schedule and play-by-play data from the `nflreadr` package, along with additional R packages such as `dplyr`, `tidyr`, `ggplot2`, `knitr`, and `geosphere`. Stadium latitude and longitude data will be used to calculate travel distances for away teams.

Steps

1. Carefully read through the full assignment instructions to ensure all expectations and requirements are understood.
2. Develop an initial analysis plan outlining the research question, methodology, and structure of the Quarto document.
3. Set up an RStudio Project to organize all files related to the analysis in a single working directory.
4. Create a new Quarto document and configure the YAML header to include relevant metadata and PDF formatting options.
5. Remove the default placeholder content and establish section headers to clearly structure the analysis.
6. Load and preprocess NFL schedule data to calculate basic home and away-team statistics, including win percentages.
7. Import NFL play-by-play data and compute game-level team performance metrics such as Expected Points Added (EPA), total yards, completion percentage, and interceptions.
8. Label teams as home or away for each game and generate comparative tables summarizing average performance in home versus away situations.
9. Collect and organize stadium latitude and longitude data, then calculate straight-line travel distances for away teams using the `geosphere` package.
10. Create example tables to illustrate differences in travel distance between selected team matchups.
11. Combine travel distance data with away-team performance metrics at the game level and create visualizations to examine the relationship between distance traveled and performance.
12. Interpret the results of the visualizations and discuss whether travel distance helps explain away-team performance differences.
13. Add a concluding section summarizing key findings, limitations, and potential directions for future analysis.
14. Include a code appendix that displays all analysis code while keeping code output hidden in the main body of the document.
15. Review the completed Quarto document for clarity, accuracy, and formatting before final submission.